ngee ann polytechnic

School of InfoComm Technology

Data Visualisation & Story Telling

Specialist Diploma in Data Analytics

Jan 2025 Semester

ASSIGNMENT 1
(Individual Assignment)

Submission Deadline:

23rd February 2025 (Sunday), 2359 hrs

Tutorial Group :	
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Penalty for late submission:
10% of the marks will be deducted every calendar day after the deadline.
NO submission will be accepted after **23 February 2025 (Sunday), 2359 hrs**

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1. Project Objectives

The aim of this project is to provide a comparative analysis of City Hotel and Resort Hotel, determine which hotel is performing better than the other, and suggest ways to increase both the volume of customers and revenue.

1.1 At which month does the Average Daily Rate (ADR) reaches its peak for City and Resort Hotels?

Understanding the trends in Average Daily Rate (ADR) helps hotel management implement strategic pricing and maximize revenue during high-demand periods. By identifying when ADR is highest, the hotel can optimize promotions, staffing, and service offerings to enhance profitability.

1.2 In which countries is this hotel brand the most popular among travelers?

Identifying the most popular hotel outlets helps to refine marketing and business strategies. It allows the management to understand which markets are performing well and which outlets might need rebranding or improvement to attract more guests. This metric is important for future decisions on the expansion of hotel outlets.

1.3 Which hotel experiences a higher volume of customer bookings?

This metric allows the hotel management to evaluate if their packages, deals and services are attractive to consumers and may provide insights into customer preferences. It gives the management a competitive advantage in identifying any branding and expansion strategies that need to be refined.

1.4 What is the optimal length of the stay in City and Resort hotels to get the best average daily rate?

Understanding this metric enables management to determine the ideal length of stay that maximizes revenue and customer satisfaction. It also allows management to adjust pricing strategies, promotions and minimum stay requirements to optimize profitability.

1.5 Which hotel has the higher Average Daily Rate (ADR)?

Identifying which hotel has the higher ADR helps the management to understand market positioning, profitability and pricing efficiency. A higher ADR indicates stronger pricing power and premium services and at a high-demand location. Having this metric allows management to understand if pricing adjustment is required.

2. Data Preparation

2.1 Data Preparation Introduction

The data is generally clean. To further clean and tidy up the data, it was loaded into power query to have a quick check for errors and null values.

2.2 Data Cleaning Process

As shown in Figure 1 below, the first two rows indicate zero stays during weekends and weeknights, which is illogical as at least one night is required to book a hotel room.

stays_in_weekend_nights	stays_in_week_nights
0	0
0	0
0	1
0	1

Figure 1

Believing this to be an electronic recording error, an advanced filter was introduced to remove rows with zero values for both weekend and weeknight stays, as shown in Figure 2 below.

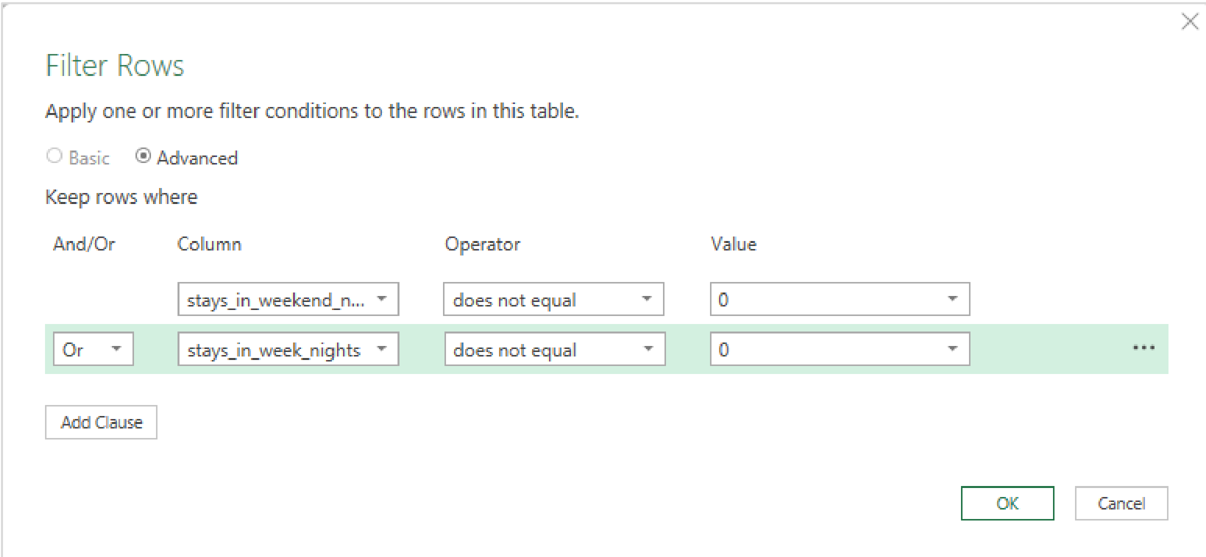

Figure 2

After closing and loading the dataset onto a new Excel sheet within the same file, the Queries & Connections tab indicated that there were 4 errors in the dataset, as shown in Figure 3.

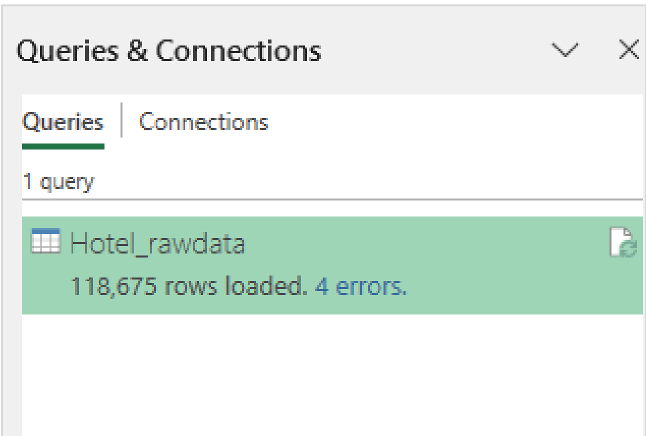

Figure 3

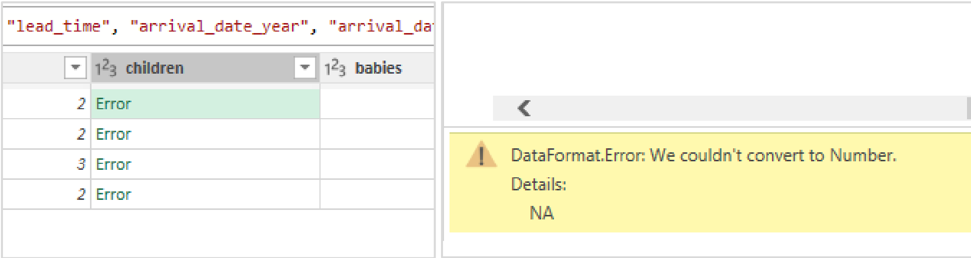

Figure 4

Upon investigating these errors, it was found that the 'children' column contained the value 'NA', which is a string instead of a number, as shown in Figure 4.

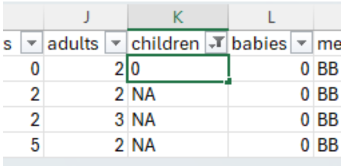

Figure 5

To fix these errors, the string 'NA' was substituted with the numerical value 0, as shown in Figure 5.

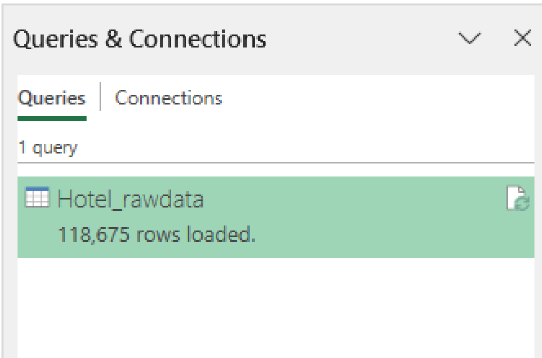

Figure 6

After resolving the errors, the dataset was reloaded and saved in a new Excel workbook named 'hotel_bookings_cleaned'.

3 Exploratory Data Analysis & Visualization

3.1 Average Daily Rate (ADR) Comparison

Figure 7 below presents a tree map showing the Average Daily Rate (ADR) for City and Resort Hotels over various months from 2015 to 2017. The map employs color intensity and box size to represent ADR values, with the darkest color and largest box indicating the highest ADR, and the lightest color and smallest box indicating the lowest ADR.

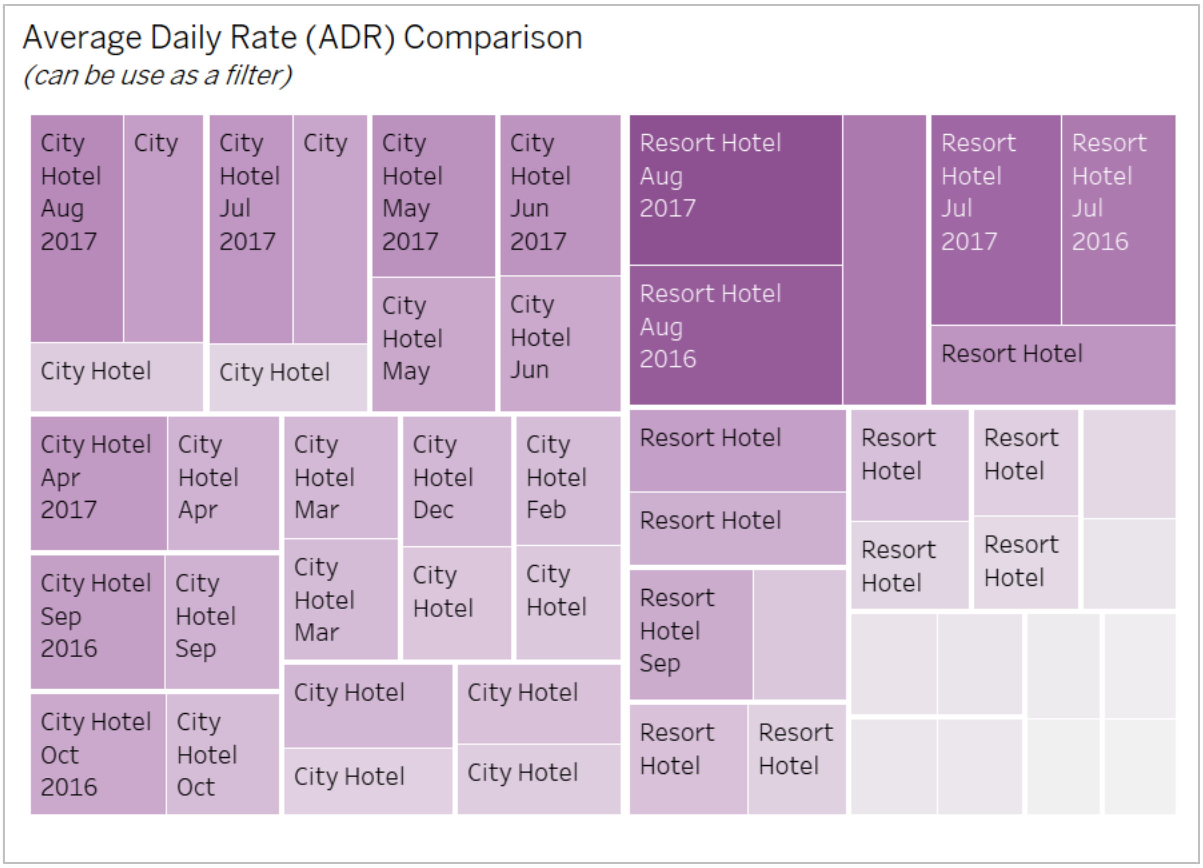

Figure 7. ADR Tree Map

The tree map was selected to address the first exploratory question: In which month does the Average Daily Rate (ADR) peak for City and Resort Hotels? This visualization makes it easy to identify the month and year when each hotel has the highest ADR, as indicated by the largest box size and the darkest color. Figure 7 indicates that the Resort Hotel had a higher Average Daily Rate (ADR) compared to the City Hotel in August 2016 and 2017.

3.2 Crowd Sizes Across Different Months

Figure 8 below illustrates the number of customers who checked into the hotels across different months from 2015 to 2017. The histogram displays the distribution of customers with two distinct colored stacked bars: the green portion represents the number of customers staying at Resort Hotels, while the purple portion indicates the number of customers staying at City Hotels. A new calculated field was created in Tableau by adding values from the adult, children, and babies columns in each row. This calculated field was named “crowd calculation”.

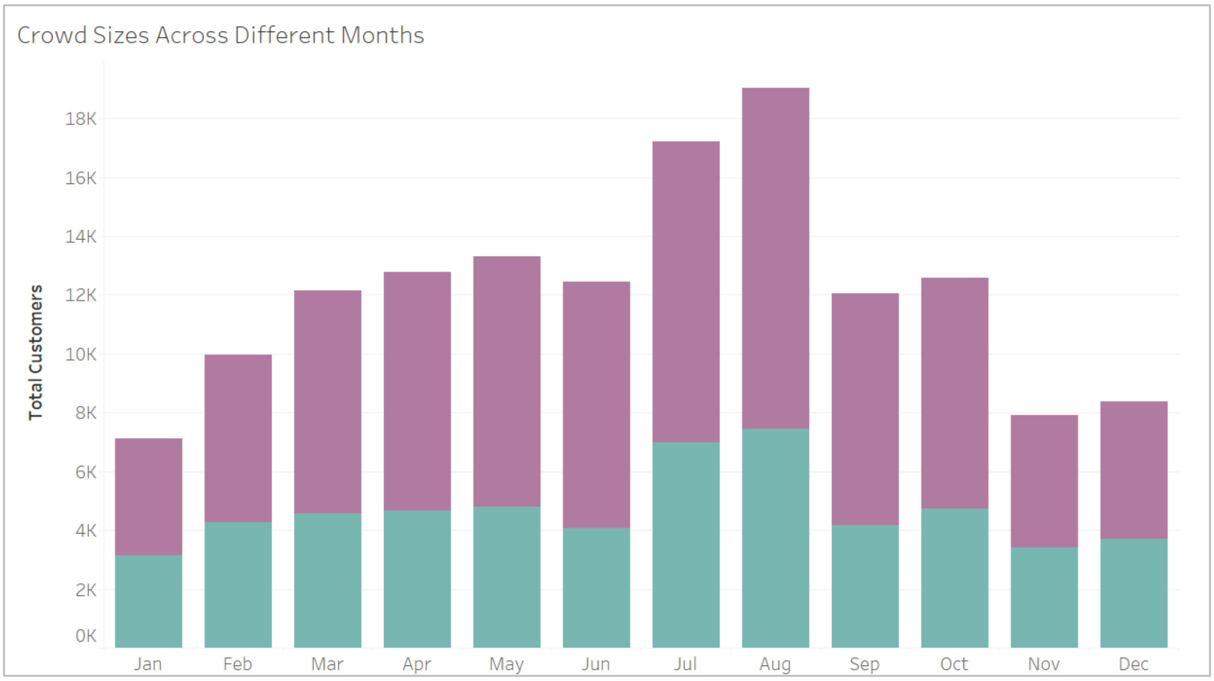

Figure 8. Histogram of Total Customers Across Different Months

The histogram was chosen to address the third exploratory question: Which hotel experiences a higher volume of customer bookings? This is because histograms effectively display crowd sizes across different months, making it easier to compare these sizes over time. The use of stacked bars allows for a clear distinction between the two hotels, enabling a side-by-side comparison of their crowd sizes through the bar heights. Figure 8 clearly shows that the City Hotel had larger crowd sizes than Resort Hotel across all months from 2015 to 2017. Additionally, this visualization highlights that both hotels experienced larger crowds in July and August compared to other months.

3.3 Optimal Length of Stay for Best Average ADR

Figure 9 below presents a table comparing City Hotel and Resort Hotel in terms of average ADR, average stays on weeknights, and average stays on weekend nights. These averages help identify booking patterns for both the City Hotel and Resort Hotel.

Optimal Length of Stay for Best Avg. ADR

	City Hotel	Resort Hotel
Avg. Adr	105.7	95.9
Avg. Stays In Week Nights	2.2	3.2
Avg. Stays In Weekend Nights	0.8	1.2

Figure 9. Table of Optimal Length of Stay for Best Average ADR

A table was selected to address the fourth and fifth exploratory questions:

1. What is the optimal length of the stay in City and Resort hotels to get the best average daily rate?
2. Which hotel has the higher Average Daily Rate (ADR)?

Since there are only three elements to compare, a table is an effective way to present the values clearly and concisely. This straightforward format allows for easy interpretation, making the table an efficient method to showcase the values. The table shows that City Hotel has a higher Average Daily Rate (ADR) than Resort Hotel. It also shows that the customers having shorter stays on both weeknights and weekend nights at City Hotel than Resort Hotel.

3.4 Customer Distribution by Location

Figure 10 below illustrates the origins of the customers.

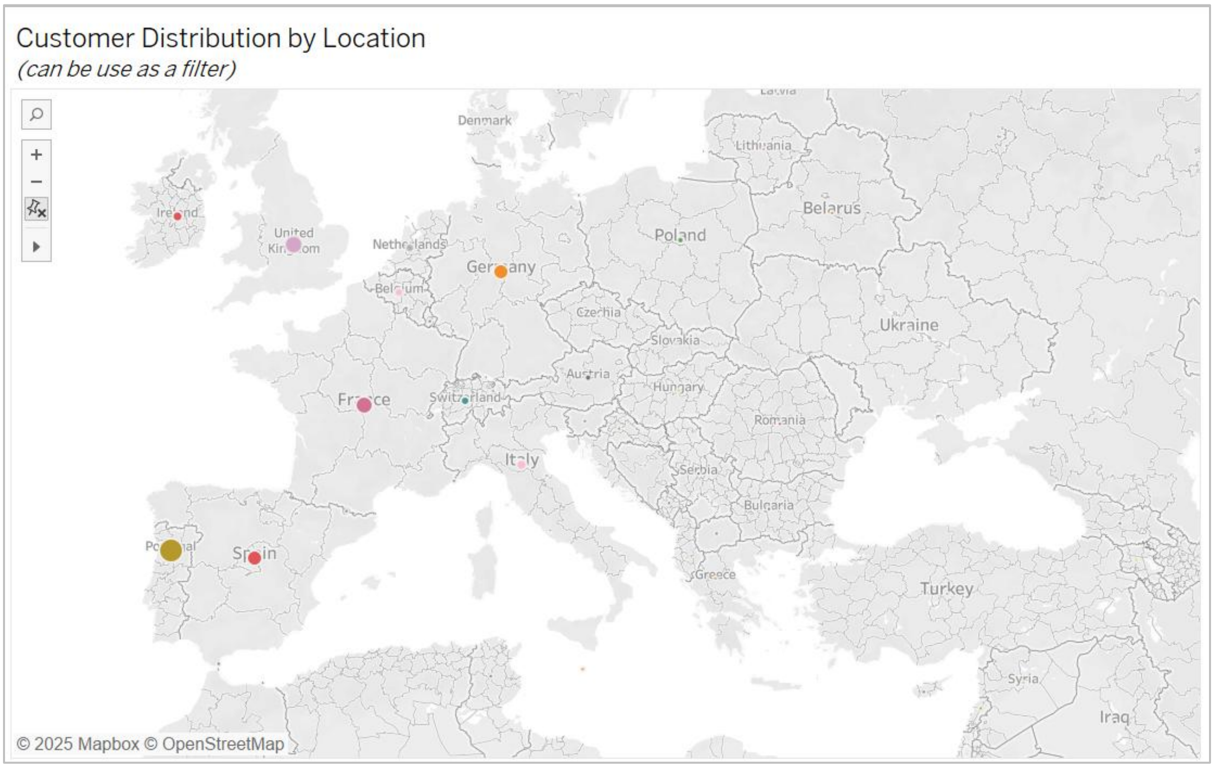

Figure 10. Spot Map of Customer Distribution by Location

A spot map was chosen to address the second exploratory question: In which countries is this hotel brand the most popular among travellers? This visualization effectively displays customer locations, making it easy to identify high-density areas and distribution gaps. It also helps to spot trends and clusters, providing a quick visual summary of the customers' countries of origin.

Figure 10 reveals that most of the customers are from western countries, specifically Germany, France, Spain, and Portugal, for both City Hotel and Resort Hotel. This indicates that both hotel brands are quite well-known in western countries.

4 Dashboard

The dashboard offers hotel management a comprehensive overview to evaluate their performance regarding revenue and occupancy rates. The dashboard is also able to provide comparisons with their competitor (i.e. City/Resort Hotel). Management can view the performance of City Hotel or Resort Hotel from 2015 to 2017 by selecting the hotel name in the top right corner of the dashboard, as illustrated in Figure 11 below.

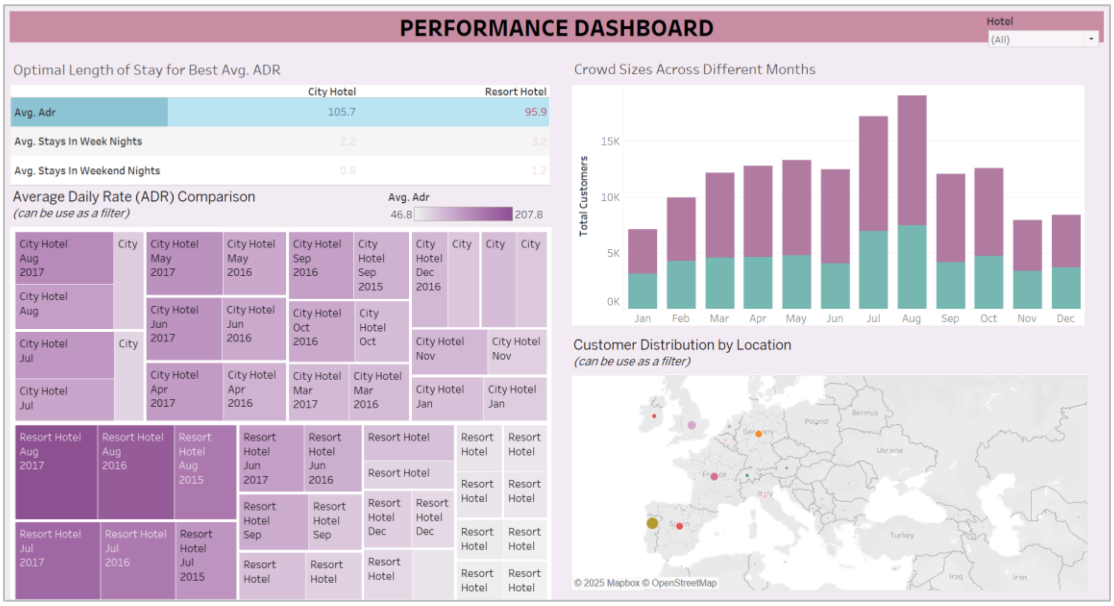

Figure 11. Performance Dashboard

Management can utilize this dashboard to analyze specific customer groups. For instance, to gauge their popularity among French visitors, they can click on "France" on the spot map. This action will filter the number of French guests, their stay patterns, the average ADR they booked, and highlight the month with the highest ADR for French visitors, as illustrated in Figure 12 below.

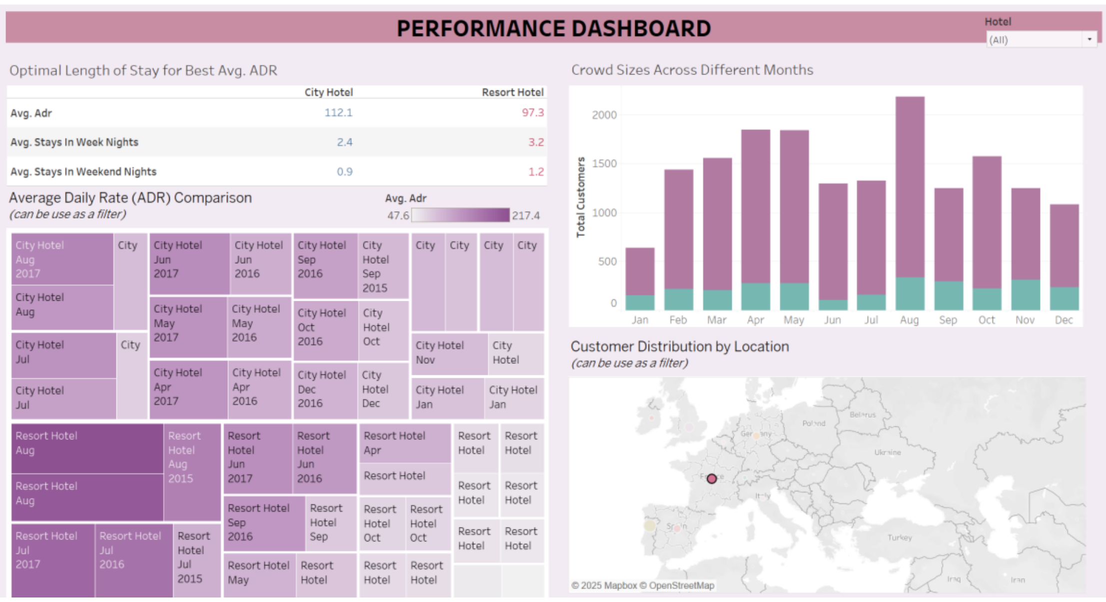

Figure 12. Performance Dashboard with "France" Filter Applied

Management can also view occupancy rates, customer distribution, and stay patterns for a specific month, year, and hotel by clicking on the specific box in the tree map, as shown in Figure 13.

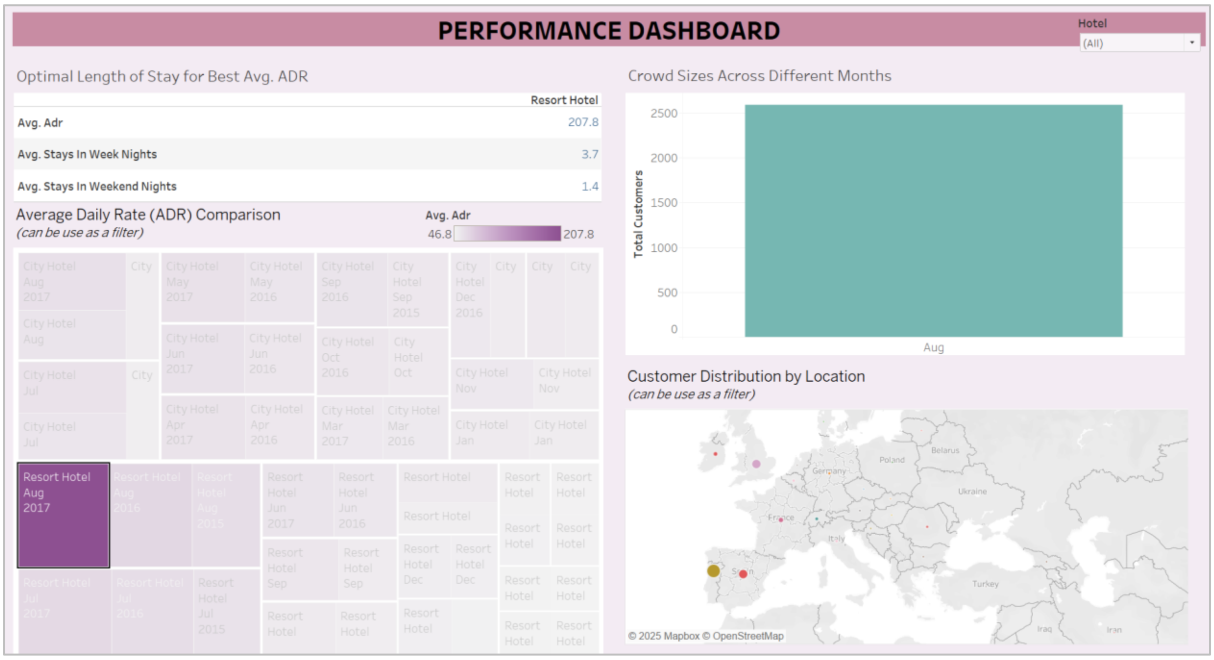

Figure 13. Performance Dashboard with "Resort Hotel Aug 2017" Filter Applied